The major, course by course

B.S. COMPUTER SCIENCE · SCU BULLETIN 2026–27

→ is a prerequisite for

→ feeds a course offered only in alternate years

offered in alternate years (department schedule, May 2026)

computer science course

Hover a course to light up what it needs and what it unlocks; hover an emphasis to see its courses. CSEN and ECEN prerequisites are not drawn.

LOWER DIVISION · all required

MATH 11
Calculus I
Prereq: MATH 9 / CRE ≥ 76 / AP

MATH 12
Calculus II
Prereq: 11

MATH 13
Calculus III
Prereq: 12

MATH 14
Calculus IV
Prereq: 13

MATH 51
Discrete Mathematics
no prerequisite · also CSEN 19

MATH 53
Linear Algebra
Prereq: 13

CSCI 10
Intro to Computer Science
Python · weekly lab

CSCI 60
C++ & Object-Oriented Programming
Prereq: CSCI 10

CSCI 61
Data Structures
Prereq: CSCI 60

CSCI 62
Advanced Programming
Prereq: CSCI 60, 61

PHYS 31 · CHEM 11 · ENVS 21 · 23
One science course
any one of the four

CSEN 20 · 20L · ECEN 21 · 21L
Embedded systems · logic design
School of Engineering

REQUIRED UPPER DIVISION

MATH 122
Probability & Statistics I
Pre/co-req: 14

CSCI 161
Automata & Languages
Prereq: MATH 51

CSCI 163
Theory of Algorithms
Prereq: MATH 51, CSCI 61 · also CSEN 179

CSEN 177 · 177L
Operating Systems
School of Engineering

MINOR IN COMPUTER SCIENCE

MINOR
Computer Science
CSCI 10, 60, 61, 62; MATH 51; CSEN 20 & 20L; two upper-division CSCI and one CSCI or CSEN

ALGORITHMS & COMPLEXITY

CSCI 162
Computational Complexity
Prereq: MATH 51

CSCI 164
Advanced Theory of Algorithms
Prereq: CSCI 163

CSCI 165
Data Privacy & Algorithmic Fairness
Prereq: MATH 51, CSCI 10 · by request

CSCI 146
Optimization I
Prereq: CSCI 10, MATH 14, 53 · also MATH 146

CSCI 147
Optimization II
Prereq: MATH 122, CSCI 146 · also MATH 147

DATA SCIENCE

CSCI 183
Data Science & Machine Learning
Prereq: CSCI 62, MATH 53, 122

CSCI 184
Advanced Machine Learning
Prereq: CSCI 183

CSCI 185
Data Mining
Prereq: CSCI 62, MATH 53, 122

CSCI 127
Computer Simulation
Prereq: CSCI 10 (MATH 122 recommended) · by request

CSCI 166
Numerical Analysis
Prereq: CSCI 10, MATH 53 · also MATH 166

SOFTWARE

CSCI 169
Programming Languages
Prereq: CSCI 62, MATH 51

CSCI 187
Design & Management of Software
Prereq: CSCI 62

CSCI 168
Computer Graphics
Prereq: CSCI 62, MATH 13

SECURITY

MATH 178
Cryptography
Prereq: MATH 51

CSCI 180
Computer Security
Prereq: CSEN 20, CSCI 62

CSCI 181
Applied Cryptography
Prereq: MATH 178, CSCI 62

MATH ELECTIVES · *named by the emphases*

MATH 101
Survey of Geometry
Prereq: 13 · algorithms emphasis

MATH 175
Theory of Numbers
Prereq: MATH 52 (not in the CS major) · algorithms, security

MATH 176
Combinatorics
Prereq: 51 · algorithms emphasis

MATH 177
Graph Theory
Prereq: 51 · algorithms emphasis

MATH 123
Probability & Statistics II
Prereq: 122; 53 pre/co · data science emphasis

CSEN ELECTIVES
146, 152, 161, 166, 169, 178
School of Engineering courses named by the emphases; prerequisites not drawn

EMPHASES · *choose one · five upper-division courses*

Algorithms & Complexity
CSCI 162, 164 · two of CSCI 146, 147, 165, 181, MATH 101, 175, 176, 177, 178 · one more from that list or any upper-division CSCI/CSEN

Data Science
CSCI 183, 184, 185 · two of CSCI 127, 146, 147, 164, 166, MATH 123, CSEN 166, or any upper-division CSCI/CSEN

Security
MATH 178, CSCI 180, 181 · two of MATH 175, CSEN 152, 161, 146, or any upper-division CSCI/CSEN (not CSEN 150)

Software
CSCI 169, 187, CSEN 146 · one of CSCI 183, 180, 168 or any upper-division CSCI · one of CSCI 183, 180, 168, CSEN 161, 178 or any upper-division CSCI/CSEN

For every emphasis
Five 4- or 5-unit upper-division courses. Decide by the end of junior year; only one emphasis. Data Science cannot be the emphasis for both a math and a CS major.